

ADVANCED NETWORK SECURITY AND ARCHITECTURES

METI

Report Module 1 [EN]

Authors:

José Oliveira (103771)

Afonso Mendes (116024)

Tiago Videira (116069)

jose.novais.oliveira@tecnico.ulisboa.pt

afonsofmenDES@tecnico.ulisboa.pt

tiago.g.videira@tecnico.ulisboa.pt

Group 3

Contents

1	Introduction	2
2	Network attacks and countermeasures	3
2.1	Objective	3
2.2	MAC Table Overflow	4
2.2.1	Attack and Countermeasure	4
2.2.2	Switch Comparison	4
2.2.3	Feasibility	5
2.3	DHCP spoofing	5
2.3.1	Attack and Countermeasure	5
2.3.2	AI Prediction	6
2.3.3	Switch Comparison	6
2.3.4	Feasibility	6
2.4	ARP poisoning (MitM)	7
2.4.1	Network Topology	7
2.4.2	Attack and Countermeasure	7
2.4.3	Defense Statistics	8
2.4.4	AI Configuration vs Experimental Setup	9
2.4.5	Switch Comparison	9
2.4.6	Feasibility	9
2.5	DNS spoofing	9
2.6	RIP poisoning	9
3	Firewalls	10
4	AAA	11

1 Introduction

FAZER APENAS NO FIM DOS 3 LABS

2 Network attacks and countermeasures

2.1 Objective

The main objective of this laboratory work is to study and experimentally evaluate a set of common network attacks and their corresponding countermeasures in a controlled environment using GNS3.

In particular, the following attacks are addressed:

- **MAC table overflow:** to understand how a switch can be forced to behave as a hub by saturating its MAC address table, and to evaluate mitigation mechanisms such as port security.
- **DHCP spoofing:** to demonstrate how a rogue DHCP server can provide malicious network configurations to clients, and to analyze the effectiveness of DHCP snooping as a countermeasure.
- **ARP poisoning (Man-in-the-Middle):** to perform a MitM attack by manipulating ARP tables, and to study the use of Dynamic ARP Inspection (DAI) for protection.
- **DNS spoofing:** to analyze how a victim can be redirected to a fake web server through DNS manipulation combined with ARP poisoning.
- **RIP poisoning:** to investigate how routing tables can be corrupted using forged RIP messages, leading to traffic redirection, and to explore authentication-based defenses.

Additionally, this work aims to:

- Analyze network behavior through packet captures using Wireshark;
- Interpret device-level information using Cisco IOS commands;
- Evaluate the practical feasibility and limitations of each attack;
- Compare experimental results with theoretical expectations.

2.2 MAC Table Overflow

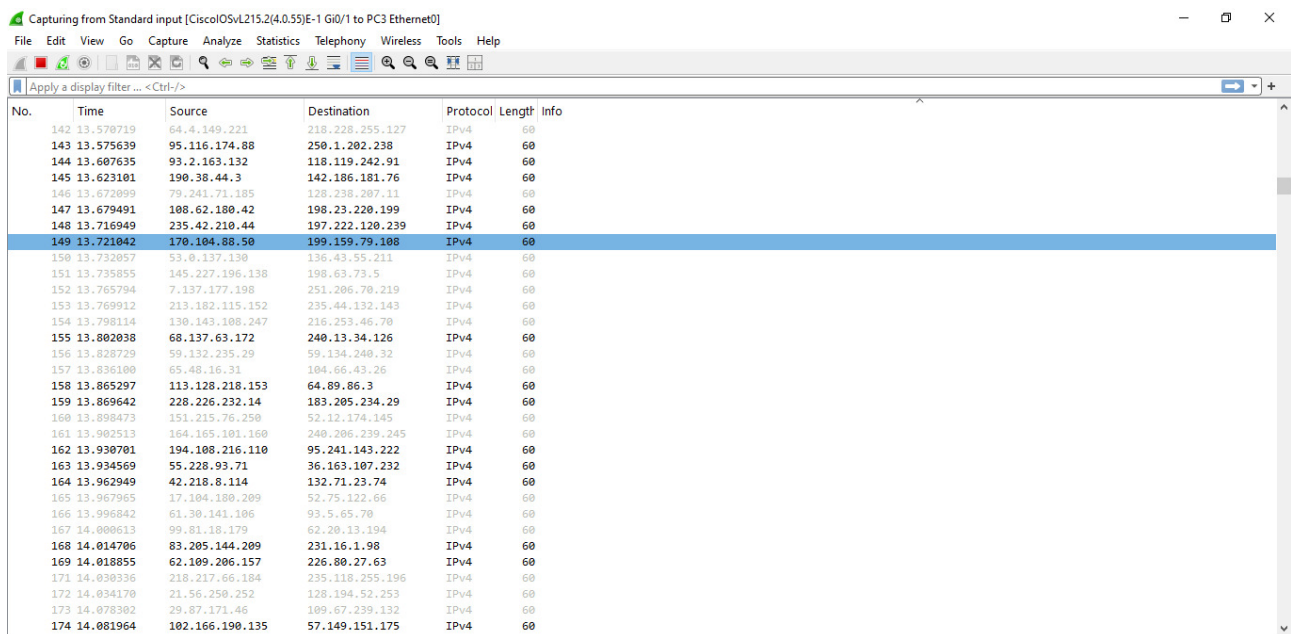
2.2.1 Attack and Countermeasure

The MAC table overflow attack was performed using the `camov.py` script, generating a large number of frames with different spoofed MAC addresses.

As observed in Wireshark (Figure 1), a high volume of traffic with varying source addresses was captured, indicating the flooding behavior of the attack.

The switch behavior was analyzed using `show mac address-table`.

To mitigate the attack, port security was configured, limiting the number of MAC addresses per port and preventing the insertion of multiple spoofed entries.



No.	Time	Source	Destination	Protocol	Length	Info
142	13.570719	64.4.149.221	218.228.255.127	IPv4	60	
143	13.575639	95.116.174.88	250.1.202.238	IPv4	60	
144	13.607635	93.2.163.132	118.119.242.91	IPv4	60	
145	13.623101	190.38.44.3	142.186.181.76	IPv4	60	
146	13.672099	79.241.71.185	128.238.207.11	IPv4	60	
147	13.679491	108.62.180.42	198.23.220.199	IPv4	60	
148	13.716949	235.42.210.44	197.222.120.239	IPv4	60	
149	13.721042	170.104.88.50	199.159.79.108	IPv4	60	
150	13.732057	33.0.137.130	136.43.59.211	IPv4	60	
151	13.735855	145.227.196.138	198.63.73.5	IPv4	60	
152	13.765794	7.137.177.198	251.206.70.219	IPv4	60	
153	13.769912	213.182.115.152	235.44.132.143	IPv4	60	
154	13.798114	130.143.108.247	216.253.46.70	IPv4	60	
155	13.802038	68.137.63.172	240.13.34.126	IPv4	60	
156	13.828729	59.132.235.29	59.134.240.32	IPv4	60	
157	13.836100	65.48.16.31	104.66.43.26	IPv4	60	
158	13.865297	113.128.218.153	64.89.86.3	IPv4	60	
159	13.869642	228.226.232.14	183.205.234.29	IPv4	60	
160	13.898473	151.215.76.250	52.12.174.145	IPv4	60	
161	13.902513	164.165.101.160	240.206.230.245	IPv4	60	
162	13.930701	194.108.216.110	95.241.143.222	IPv4	60	
163	13.934569	55.228.93.71	36.163.107.232	IPv4	60	
164	13.962949	42.218.8.114	132.71.23.74	IPv4	60	
165	13.967965	17.104.180.209	52.75.122.66	IPv4	60	
166	13.996842	61.30.141.106	93.5.65.70	IPv4	60	
167	14.000613	99.81.18.179	62.20.13.194	IPv4	60	
168	14.014706	83.205.144.209	231.16.1.98	IPv4	60	
169	14.018855	62.109.206.157	226.80.27.63	IPv4	60	
171	14.030336	218.217.66.104	235.118.255.196	IPv4	60	
172	14.034170	21.56.250.252	128.194.52.253	IPv4	60	
173	14.078302	29.87.171.46	109.67.239.132	IPv4	60	
174	14.081964	102.166.190.135	57.149.151.175	IPv4	60	

Figure 1: Wireshark capture during MAC table overflow attack

2.2.2 Switch Comparison

Cisco Catalyst 2960 supports port security, allowing the configuration of a maximum number of MAC addresses per port and defining actions when violations occur (e.g., restrict or shutdown). This makes it possible to prevent MAC table overflow attacks.

Link: www.cisco.com/c/en/us/products/switches/catalyst-2960-series-switches/index.html

In contrast, the TP-Link TL-SF1005D is an unmanaged switch that does not support port security or any traffic control mechanisms, as it lacks a configuration interface. Therefore, it cannot prevent MAC flooding attacks.

Link: www.tp-link.com/en/home-networking/soho-switch/tl-sf1005d/

2.2.3 Feasibility

The objective of this attack is to flood the switch MAC address table, forcing the switch to behave like a hub. In that state, frames may be forwarded to all ports, allowing the attacker to listen to conversations between other hosts.

However, in practice, the attack may be difficult to execute successfully on modern switches because their MAC address tables are large. Additionally, the high amount of generated traffic can cause network instability or a denial-of-service condition, instead of allowing the attacker to reliably capture useful traffic.

Therefore, although the attack is theoretically possible, its practical feasibility is limited and depends on the switch capacity, configuration, and security features.

2.3 DHCP spoofing

2.3.1 Attack and Countermeasure

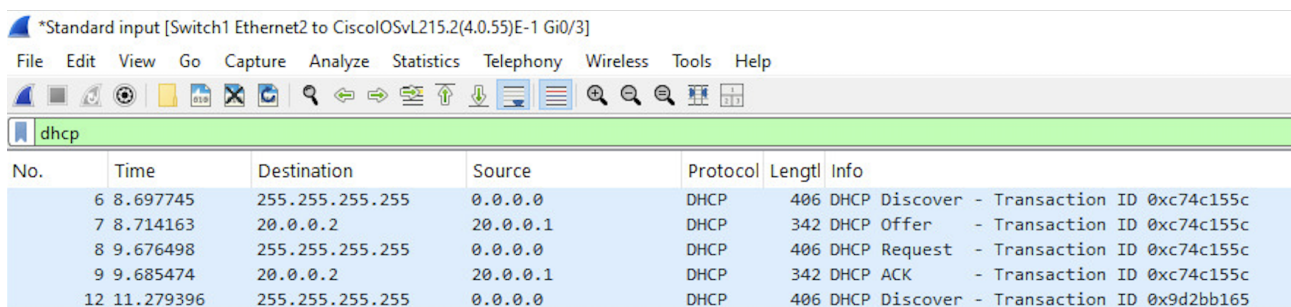
The DHCP spoofing attack was performed by introducing a rogue DHCP server that responds to DHCP Discover messages with malicious configuration parameters.

As observed in Wireshark (Figure 2), multiple DHCP Offer messages were received for the same transaction ID, indicating a race condition between the legitimate and the rogue DHCP servers. The victim accepts the first Offer received, which may result in a malicious gateway configuration provided by the attacker.

20	20.480048	20.0.0.100	20.0.0.200	DHCP	338 DHCP Offer	- Transaction ID 0x2c5ff044
21	20.482689	20.0.0.2	20.0.0.1	DHCP	342 DHCP Offer	- Transaction ID 0x2c5ff044
23	21.447930	255.255.255.255	0.0.0.0	DHCP	406 DHCP Request	- Transaction ID 0x2c5ff044
24	21.493701	20.0.0.100	20.0.0.200	DHCP	338 DHCP ACK	- Transaction ID 0x2c5ff044

Figure 2: Wireshark capture showing multiple DHCP Offer messages (attack)

After enabling DHCP snooping and configuring the port connected to the legitimate DHCP server as trusted, only one DHCP Offer was observed (Figure 3). The rogue DHCP responses were blocked by the switch, and the client received a valid configuration. This confirms that DHCP snooping effectively mitigates the attack.



The image shows a Wireshark capture window titled '*Standard input [Switch1 Ethernet2 to CiscoIOSvL215.2(4.0.55)E-1 Gi0/3]'. The packet list pane shows a DHCP transaction. The packet details pane shows the DHCP Offer message. The packet bytes pane shows the raw data of the DHCP Offer message.

No.	Time	Destination	Source	Protocol	Length	Info
6	8.697745	255.255.255.255	0.0.0.0	DHCP	406	DHCP Discover - Transaction ID 0xc74c155c
7	8.714163	20.0.0.2	20.0.0.1	DHCP	342	DHCP Offer - Transaction ID 0xc74c155c
8	9.676498	255.255.255.255	0.0.0.0	DHCP	406	DHCP Request - Transaction ID 0xc74c155c
9	9.685474	20.0.0.2	20.0.0.1	DHCP	342	DHCP ACK - Transaction ID 0xc74c155c
12	11.279396	255.255.255.255	0.0.0.0	DHCP	406	DHCP Discover - Transaction ID 0x9d2bb165

Figure 3: Wireshark capture with DHCP snooping enabled (only legitimate Offer)

2.3.2 AI Prediction

The AI predicted that the victim closer to the attacker would be more likely to be compromised, since the rogue DHCP Offer would arrive first. This prediction matched the experimental results, where the victim receiving the first Offer obtained the malicious configuration.

2.3.3 Switch Comparison

Cisco Catalyst 2960 supports DHCP snooping, allowing ports to be classified as trusted or untrusted and blocking DHCP server messages from untrusted ports.

Link: www.cisco.com/c/en/us/products/switches/catalyst-2960-series-switches/index.html

The TP-Link TL-SF1005D does not support DHCP snooping, as it is an unmanaged switch without configuration capabilities or security features.

Link: www.tp-link.com/en/home-networking/soho-switch/tl-sf1005d/

2.3.4 Feasibility

The attack is feasible in networks without DHCP snooping, especially when the attacker is closer to the victim than the legitimate DHCP server. However, due to the race condition, success is not guaranteed. When DHCP snooping is enabled, the attack becomes ineffective.

2.4 ARP poisoning (MitM)

2.4.1 Network Topology

The ARP poisoning attack was performed in the network shown in Figure 4, where two hosts communicate through a switch and an attacker is connected to the same Layer 2 domain.

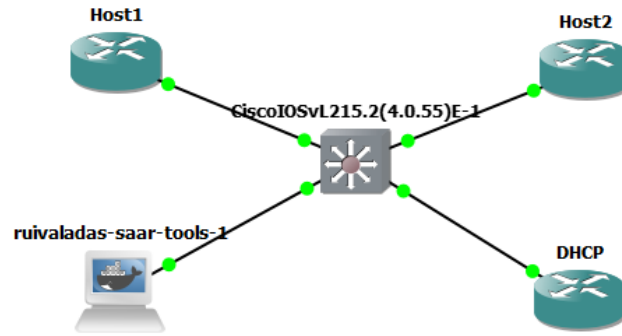


Figure 4: Network topology for ARP poisoning attack

2.4.2 Attack and Countermeasure

As observed in Wireshark (Figure 5), ARP packets indicate duplicate IP usage, confirming that the attacker is associating its MAC address with multiple IP addresses.

At the same time, ICMP traffic between the victims is still observed, which demonstrates that communication is maintained while being intercepted by the attacker, confirming a successful Man-in-the-Middle (MitM) condition.

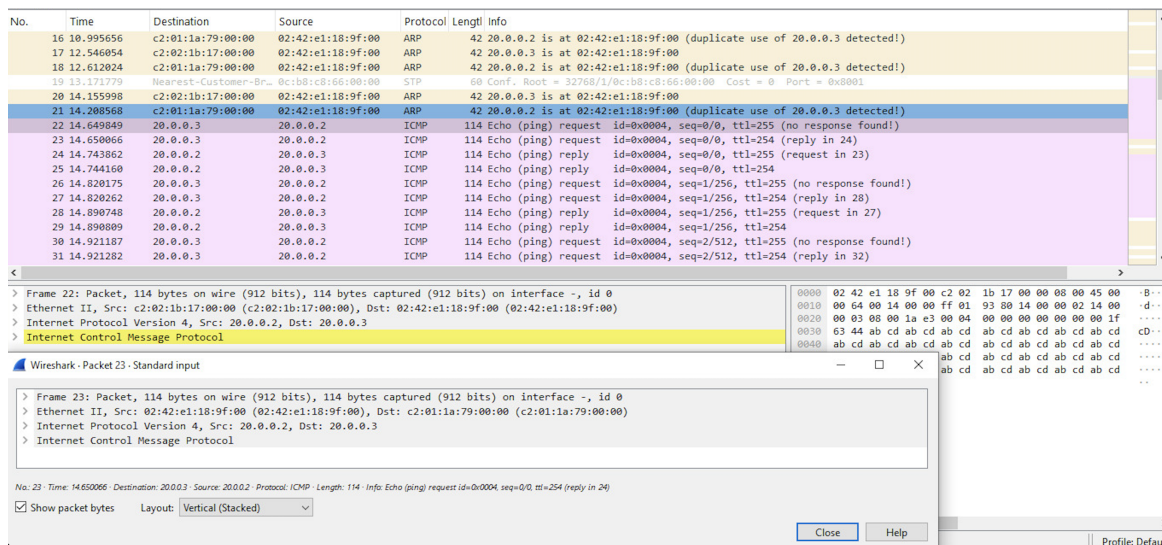


Figure 5: ARP poisoning showing duplicate IP detection and intercepted ICMP traffic

The attacker generated multiple forged ARP requests and replies, attempting to poison the ARP tables of the victims, as shown in Figure 6.

22	41.442250	Broadcast	02:42:e1:18:9f:00	ARP	42 Who has 20.0.0.3? Tell 20.0.0.2
24	43.473802	Broadcast	02:42:e1:18:9f:00	ARP	42 Who has 20.0.0.4? Tell 20.0.0.2
26	45.523090	Broadcast	02:42:e1:18:9f:00	ARP	42 Who has 20.0.0.3? Tell 20.0.0.2
28	47.548118	Broadcast	02:42:e1:18:9f:00	ARP	42 20.0.0.4 is at 02:42:e1:18:9f:00
29	47.597578	Broadcast	02:42:e1:18:9f:00	ARP	42 Who has 20.0.0.4? Tell 20.0.0.2
31	49.613918	Broadcast	02:42:e1:18:9f:00	ARP	42 20.0.0.3 is at 02:42:e1:18:9f:00
32	51.161927	Broadcast	02:42:e1:18:9f:00	ARP	42 Who has 20.0.0.3? Tell 20.0.0.2
34	53.178241	Broadcast	02:42:e1:18:9f:00	ARP	42 20.0.0.4 is at 02:42:e1:18:9f:00
35	53.226972	Broadcast	02:42:e1:18:9f:00	ARP	42 Who has 20.0.0.4? Tell 20.0.0.2
37	55.245424	Broadcast	02:42:e1:18:9f:00	ARP	42 20.0.0.3 is at 02:42:e1:18:9f:00
41	56.794383	Broadcast	02:42:e1:18:9f:00	ARP	42 Who has 20.0.0.3? Tell 20.0.0.2
43	58.809905	Broadcast	02:42:e1:18:9f:00	ARP	42 20.0.0.4 is at 02:42:e1:18:9f:00
44	58.862239	Broadcast	02:42:e1:18:9f:00	ARP	42 Who has 20.0.0.4? Tell 20.0.0.2

Figure 6: ARP packets generated by the attacker

However, the attack was mitigated using Dynamic ARP Inspection (DAI). The switch detected invalid ARP packets and blocked them, as shown in the system logs (Figure 7).

```
*Apr 29 10:34:18.724: %SW_DAI-4-DHCP_SNOOPING_DENY: 1 Invalid ARPs (Req) on Gi0/0, vlan 1.([0242.e118.9f00/20.0.0.2/0000.0000.0000/20.0.0.3/10:34:18 UTC Wed Apr 29 2026])
*Apr 29 10:34:20.876: %SW_DAI-4-DHCP_SNOOPING_DENY: 1 Invalid ARPs (Res) on Gi0/0, vlan 1.([0242.e118.9f00/20.0.0.4/0000.0000.0000/20.0.0.3/10:34:20 UTC Wed Apr 29 2026])
*Apr 29 10:34:20.877: %SW_DAI-4-DHCP_SNOOPING_DENY: 1 Invalid ARPs (Req) on Gi0/0, vlan 1.([0242.e118.9f00/20.0.0.2/0000.0000.0000/20.0.0.4/10:34:20 UTC Wed Apr 29 2026])
*Apr 29 10:34:23.095: %SW_DAI-4-DHCP_SNOOPING_DENY: 1 Invalid ARPs (Res) on Gi0/0, vlan 1.([0242.e118.9f00/20.0.0.3/0000.0000.0000/20.0.0.4/10:34:21 UTC Wed Apr 29 2026])
*Apr 29 10:34:24.097: %SW_DAI-4-DHCP_SNOOPING_DENY: 1 Invalid ARPs (Req) on Gi0/0, vlan 1.([0242.e118.9f00/20.0.0.2/0000.0000.0000/20.0.0.3/10:34:23 UTC Wed Apr 29 2026])
*Apr 29 10:34:25.279: %SW_DAI-4-DHCP_SNOOPING_DENY: 1 Invalid ARPs (Res) on Gi0/0, vlan 1.([0242.e118.9f00/20.0.0.4/0000.0000.0000/20.0.0.3/10:34:25 UTC Wed Apr 29 2026])
*Apr 29 10:34:25.280: %SW_DAI-4-DHCP_SNOOPING_DENY: 1 Invalid ARPs (Req) on Gi0/0, vlan 1.([0242.e118.9f00/20.0.0.2/0000.0000.0000/20.0.0.4/10:34:25 UTC Wed Apr 29 2026])
*Apr 29 10:34:26.622: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on GigabitEthernet0/3 (not half duplex), with DHCP FastEthernet0/0 (half duplex).
```

Figure 7: DAI log showing invalid ARP packets being denied

2.4.3 Defense Statistics

The effectiveness of DAI is confirmed by the statistics shown in Figure 8, where a high number of ARP packets were dropped by the switch.

Switch#show ip arp inspection statistics				
Vlan	Forwarded	Dropped	DHCP Drops	ACL Drops
1	6	259	259	0
Vlan	DHCP Permits	ACL Permits	Probe Permits	Source MAC Failures
1	6	0	0	0
Vlan	Dest MAC Failures	IP Validation Failures	Invalid Protocol Data	
1	0	0	0	

Figure 8: DAI statistics showing dropped ARP packets

2.4.4 AI Configuration vs Experimental Setup

The AI suggested enabling DHCP snooping and configuring Dynamic ARP Inspection on the switch, marking trusted interfaces accordingly.

This matched the experimental setup, where DHCP snooping was required for DAI to validate ARP packets. The configuration had to be adapted to the topology, particularly in defining trusted ports and VLAN configuration.

2.4.5 Switch Comparison

Cisco Catalyst 2960 supports Dynamic ARP Inspection, allowing validation of ARP packets based on DHCP snooping bindings.

Link: www.cisco.com/c/en/us/products/switches/catalyst-2960-series-switches/index.html

The TP-Link TL-SF1005D does not support DAI, as it is an unmanaged switch without security or inspection mechanisms.

Link: www.tp-link.com/en/home-networking/soho-switch/tl-sf1005d/

2.4.6 Feasibility

The attacker attempted to poison the ARP tables by sending forged ARP messages. The attack successfully created a Man-in-the-Middle condition, as observed in Wireshark. However, based on the switch logs and DAI statistics, these packets were detected and blocked.

Therefore, although the attack is feasible in networks without protection, in this setup it was not successful due to the correct configuration of Dynamic ARP Inspection.

2.5 DNS spoofing

2.6 RIP poisoning

3 Firewalls

4 AAA